

FORT RELIANCE, NT, Canada

WMO: 711600

Lat: 62.711N

Lon: 109.168W

Elev: 550

StdP: 14.41

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -42.0	(c) -37.9	(d) -50.4	(e) 0.3	(f) -41.7	(g) -46.2	(h) 0.4	(i) -37.7	(j) 20.3	(k) -3.2	(l) 18.1	(m) -4.7	(n) 3.3	(o) 210	(p) 0.759

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	76.3	60.4	72.4	58.4	69.1	57.1	62.1	72.3	60.2	69.3	58.4	66.5	5.5	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.0	73.4	64.8	56.1	68.3	63.7	54.1	63.5	62.5	28.2	72.9	26.8	69.4	25.6	66.5	69.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.3	(b) 17.0	(c) 14.3	(e) -45.8	(f) 82.2	(g) 3.6	(h) 3.8	(i) -48.4	(j) 84.9	(k) -50.5	(l) 87.0	(m) -52.6	(n) 89.2	(o) -55.2	(p) 91.9		
(4) 20.3	(5) 17.0	(6) 14.3	(7) -45.7	(8) 64.1	(9) 3.2	(10) 1.9	(11) -48.0	(12) 65.5	(13) -49.9	(14) 66.6	(15) -51.7	(16) 67.7	(17) -54.1	(18) 69.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.8	-14.7	-11.8	-1.9	16.3	36.4	51.4	59.2	56.8	45.9	30.4	9.8	-6.5
	DBStd	28.88	14.46	14.45	16.89	14.78	10.85	7.96	6.47	6.35	7.67	8.80	11.79	14.51
	HDD50	10621	2006	1731	1609	1011	434	80	6	13	166	608	1206	1751
	HDD65	15438	2471	2151	2074	1460	888	409	202	265	573	1073	1656	2216
	CDD50	691	0	0	0	0	11	123	290	222	44	1	0	0
	CDD65	34	0	0	0	0	0	2	22	10	0	0	0	0
	CDH74	218	0	0	0	0	0	30	140	48	0	0	0	0
	CDH80	30	0	0	0	0	0	2	21	7	0	0	0	0
Wind	WSAvg	6.2	5.0	5.0	5.7	6.3	6.2	5.5	5.2	5.8	7.3	8.4	8.0	5.4
Precipitation	PrecAvg	11.1	0.5	0.4	0.4	0.6	0.7	1.3	1.5	1.8	1.4	1.2	0.8	0.6
	PrecMax	15.3	1.0	1.1	1.3	1.2	1.9	4.6	3.9	3.5	3.0	2.9	1.9	1.4
	PrecMin	7.0	0.1	0.0	0.2	0.2	0.0	0.2	0.1	0.5	0.2	0.2	0.2	0.1
	PrecStd	2.2	0.2	0.3	0.3	0.3	0.4	1.0	0.9	0.9	0.6	0.6	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.9	25.9	40.8	53.3	67.8	78.0	83.4	79.0	68.1	52.6	36.8	27.6
		MCWB	23.8	24.2	35.0	43.1	51.1	59.2	62.9	62.8	56.6	46.5	34.3	26.3
	2%	DB	18.5	19.6	34.9	46.3	61.5	73.0	78.0	73.8	63.6	47.3	33.3	23.2
		MCWB	17.8	18.6	30.2	38.2	47.6	56.7	61.4	60.2	54.3	42.9	31.5	22.0
	5%	DB	12.5	14.6	29.3	41.3	55.9	68.9	73.9	70.2	59.7	44.0	30.1	18.9
		MCWB	11.7	13.5	26.1	35.3	45.0	54.8	59.5	58.9	52.3	41.1	28.6	17.8
	10%	DB	7.5	9.0	24.1	37.2	51.1	64.9	70.0	66.8	56.3	41.3	26.9	14.6
		MCWB	6.7	8.0	21.6	32.8	42.2	52.9	57.8	57.5	50.6	38.9	25.5	13.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.8	24.1	35.9	43.3	51.8	61.2	65.4	64.4	57.8	47.0	34.8	26.2
		MCDB	24.8	25.4	40.4	53.1	67.0	73.3	77.8	74.2	65.7	51.5	36.5	27.3
	2%	WB	17.5	18.6	30.6	38.5	48.2	58.1	62.6	61.9	55.1	43.3	31.6	22.1
		MCDB	18.2	19.7	34.7	45.5	60.5	69.5	73.9	70.9	61.9	46.3	33.2	23.2
	5%	WB	11.3	13.4	26.3	35.7	45.3	55.9	60.8	60.1	52.9	41.3	28.6	17.9
		MCDB	12.1	14.4	29.2	40.9	54.9	66.9	70.7	68.2	58.7	44.1	30.1	18.9
	10%	WB	6.1	8.0	21.4	33.2	42.5	53.7	58.9	58.3	51.0	38.9	25.5	13.6
		MCDB	6.9	8.9	23.8	36.9	50.3	63.8	67.6	65.3	55.8	41.3	26.9	14.6
Mean Daily Temperature Range	5% DB	MDBR	14.2	16.5	21.2	22.0	19.0	19.3	17.3	14.3	11.2	8.2	10.7	12.8
		MCDBR	15.6	18.6	23.9	22.9	25.4	25.2	22.4	19.8	16.4	10.9	9.8	14.3
		MCWBR	14.9	17.8	20.4	15.6	14.7	12.9	9.9	9.3	9.0	7.7	9.3	13.9
	5% WB	MCDBR	15.4	18.6	23.9	21.8	24.1	23.4	19.6	17.3	15.1	10.6	10.2	14.3
		MCWBR	15.0	17.9	20.1	15.0	14.3	12.9	10.0	9.0	8.9	7.7	9.8	14.0
Clear-Sky Solar Irradiance	taub	0.181	0.238	0.254	0.309	0.328	0.349	0.368	0.352	0.310	0.268	0.191	0.147	
	taud	2.026	2.211	2.275	2.185	2.383	2.423	2.413	2.447	2.554	2.404	2.018	1.942	
	Ebn at Noon	164	234	276	277	282	277	268	262	256	222	150	113	
	Edh at Noon	10	19	26	36	33	32	32	28	21	16	10	5	
All-Sky Solar Radiation	RadAvg	86	346	906	1592	1869	1881	1678	1240	735	283	90	38	
	RadStd	6	25	48	83	129	129	96	70	63	37	12	3	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+149	N/A

Nomenclature: See separate page